ECO 331

Economic History

The Late Industrializers

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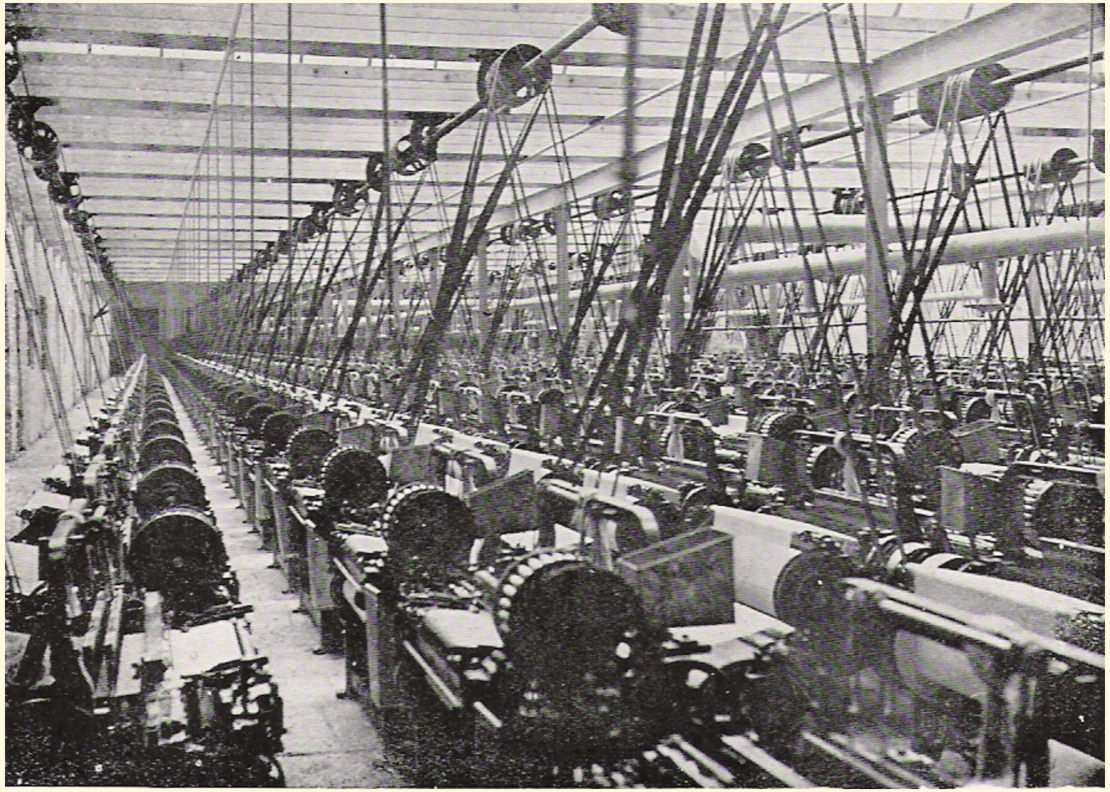
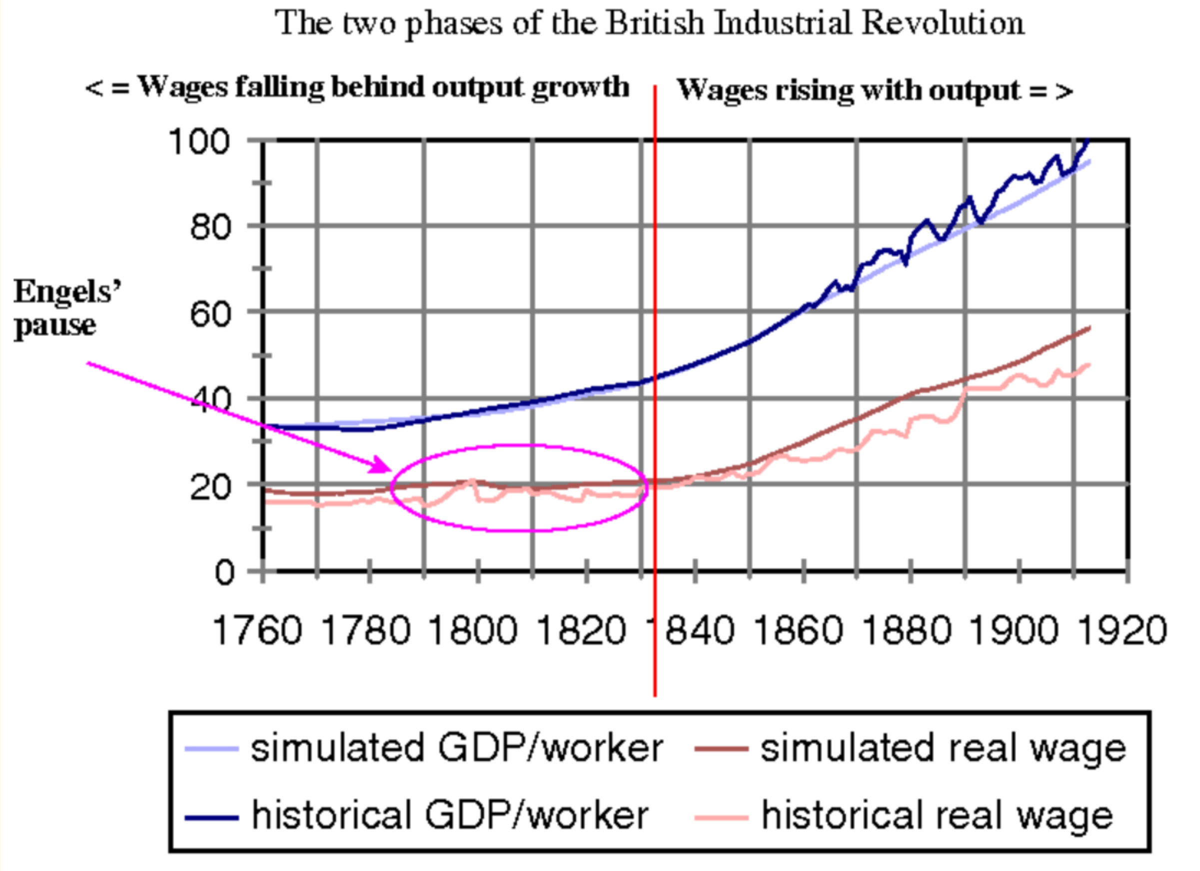
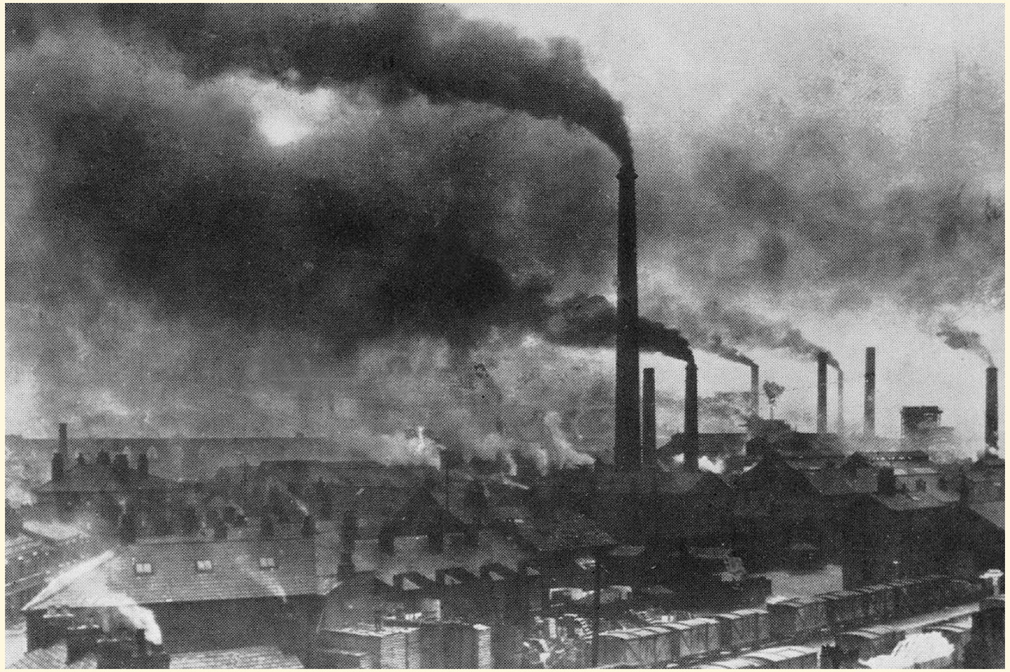


- The Industrial Revolution did not transform the lives of workers in a single generation
- British GDP per capita increased but many workers did not see major improvements in living standards, at least until after around 1840
- Allen (2009) labeled this "Engel's Pause"
- But a decisive increase in both per capita GDP and real wages by end century



Consequences of Industrialization

- Industrialization led to much pessimism about the state of the working class
- Karl Marx was the most famous proponent of the “pessimistic case”
- Urban crowding, dangerous factories, disease, child labor, pollution



Babington Macaulay's Optimism

- Macaulay was one of the first individuals to understand the long run consequences of the Industrial Revolution
- He predicted that modern economic growth would transform both England and the world
- He was proven correct

The Second Industrial Revolution (1870-1914)

- Many innovations of the First Industrial Revolution did not rely directly on scientific breakthroughs
- After 1850, innovations in chemistry, medicine, and physics had direct economic payoffs
- This was the “marriage of science and technology”. Industrial R&D.

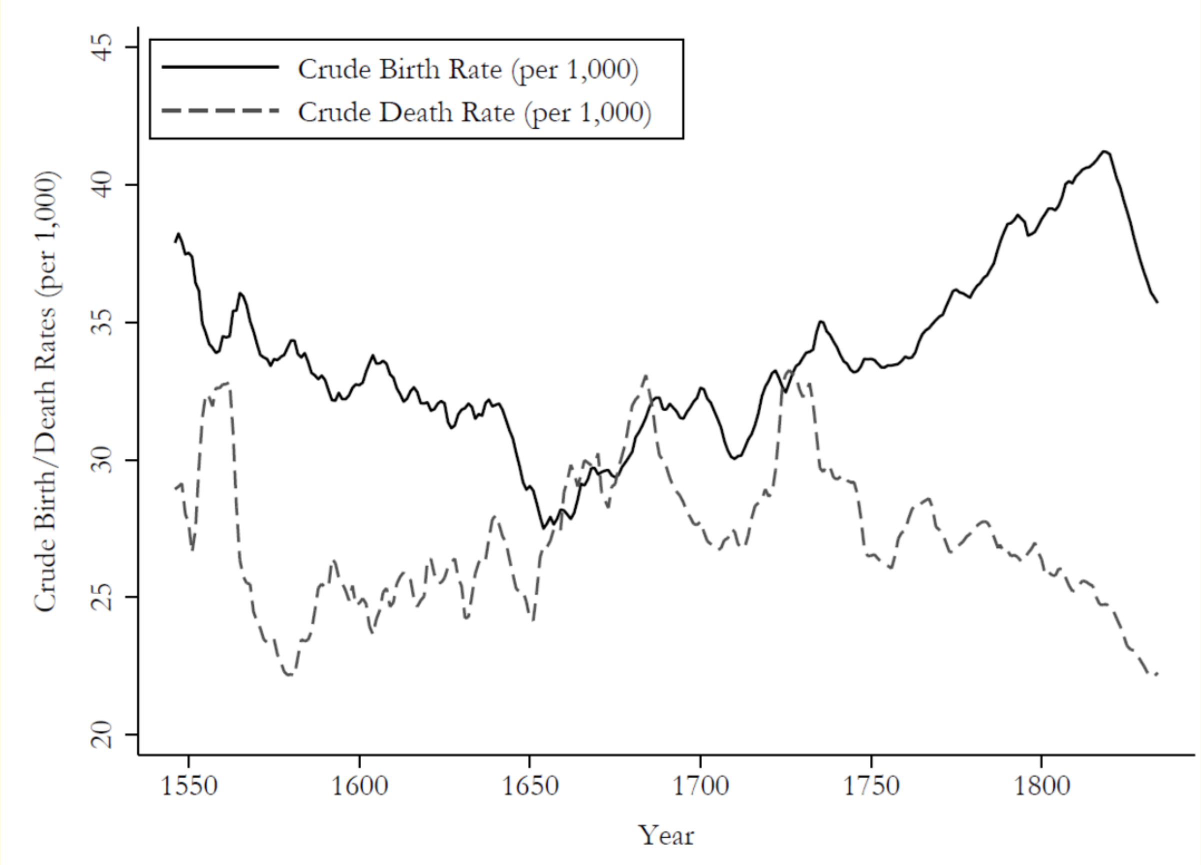
Table 9.1 Major inventions of the Second Industrial Revolution

Invented	Massively expanded or improved
Cheap steel	Electricity
– Used in construction, ships, railroad tracks	– Used in telegraph, railway, light bulb
Automobile	Railroads and diesel engine
Airplane	Telegraph (and networks)
Telephone	Vulcanized rubber
Typewriter	Anesthetics and antiseptics
Chemical dyes	Sewerage systems
Internal combustion engine	Sewing machine
Chemical fertilizers	Food canning
Synthetic plastic	Clean water supply
Dynamite	Soda-making
Sulphuric acid	Disinfectants
Aspirin	
Bicycle	
Barbed wire	
Fungicides	

Data source: Mokyr (1998).

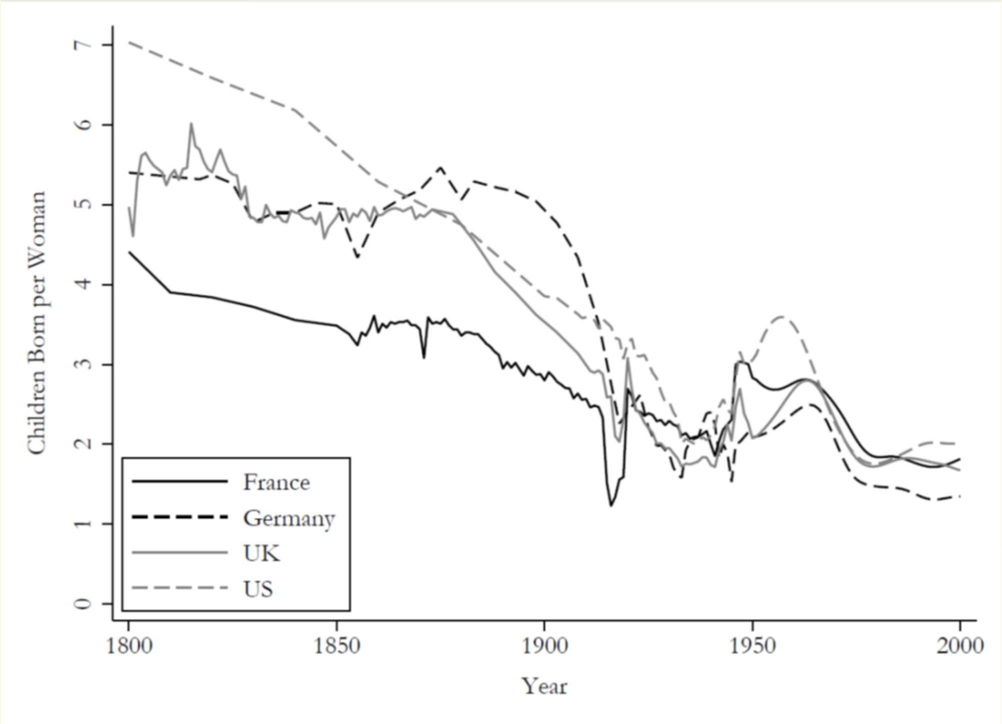
The Demographic Transition

- Beginning with France (and then later in England), people started to control fertility within marriage
- This demographic transition meant that economies stopped behaving in a Malthusian fashion
- Population growth slowed even as incomes improved
- Fertility and mortality in England/Britain, 1541-1839



The Demographic Transition

- Timing of the fertility transition is hard to explain
- Began in France in the late 18th century, before the onset of modern economic growth
- Spolaore and Wacziarg (2021) argue it diffused alongside French culture in 19th century
- As Malthusian pressures eased, resources were freed up to invest in education
- Children born per woman in
- Western Europe and the US, 1800-2000

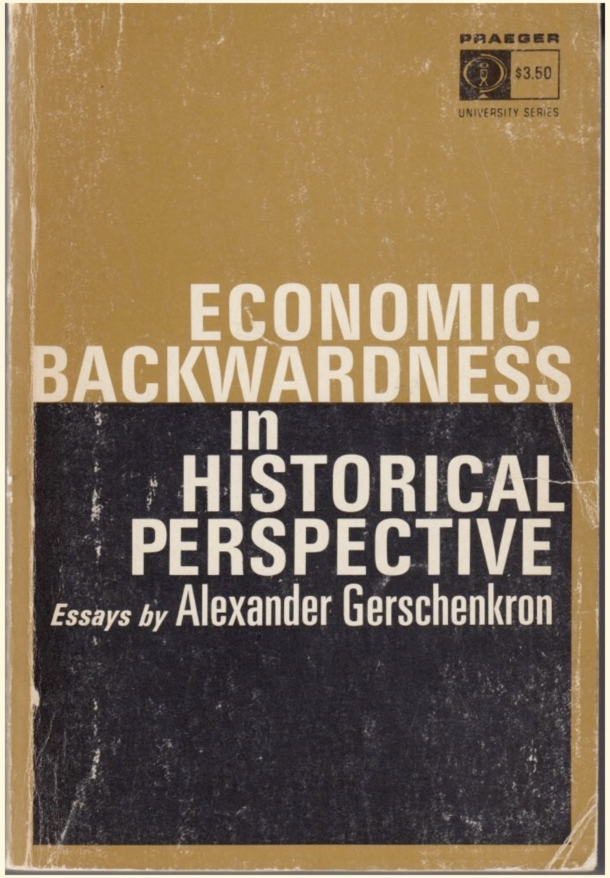


Human Capital and the Demographic Transition

- Gary Becker argued that parents choose between the quality and quantity of children
- After the demographic transition, societies invested more in education
- An educated labor force was a key complement to technologies of the Second Industrial Revolution
- Public education supported by industrialists.

Late Industrializers

- Alexander Gershenrkon: late industrializers did not have to go through same path as Britain
- Early stages of development could be leap-frogged
- Late industrializers had a backlog of advanced technologies they could adopt
- Germany harnessed capital on a larger scale (via investment banks)becoming a leader in new sectors such as chemical fertilizers (Bayer, IG Farben) and arms (Krupp)



Human Capital and the Diffusion of Industrialization

- Human capital played a critical role in the Second Industrial Revolution diffusion.
- Becker et al. (2011) find that pre-industrial levels of education (measured in 1816) in Germany predict non-textile-based industrialization in 19th

TABLE 1—EDUCATION AND INDUSTRIALIZATION IN THE FIRST PHASE OF THE INDUSTRIAL REVOLUTION

	OLS			
Dependent variable	Share of factory workers in total population 1849			
	All factories (1)	All except metal and textiles (2)	Metal factories (3)	Textile factories (4)
Years of schooling 1849[a]	0.177**	0.156***	0.059	−0.038
	(0.077)	(0.045)	(0.045)	(0.034)
Share of population < 15 years	−0.016	−0.043*	0.040	−0.013
	(0.046)	(0.026)	(0.027)	(0.014)
Share of population > 60 years	−0.092	−0.134**	−0.057	0.100***
	(0.096)	(0.063)	(0.048)	(0.038)
County area (in 1,000 km²)	−0.011***	−0.004***	−0.005**	−0.002**
	(0.002)	(0.001)	(0.002)	(0.001)
Constant	0.028	0.028**	−0.005	0.005
	(0.020)	(0.012)	(0.010)	(0.006)
Observations	334	334	334	334
R^2	0.103	0.140	0.035	0.063

The Economic Rise of the USA

- The United States became the world's leading economy by the end of the 19th century

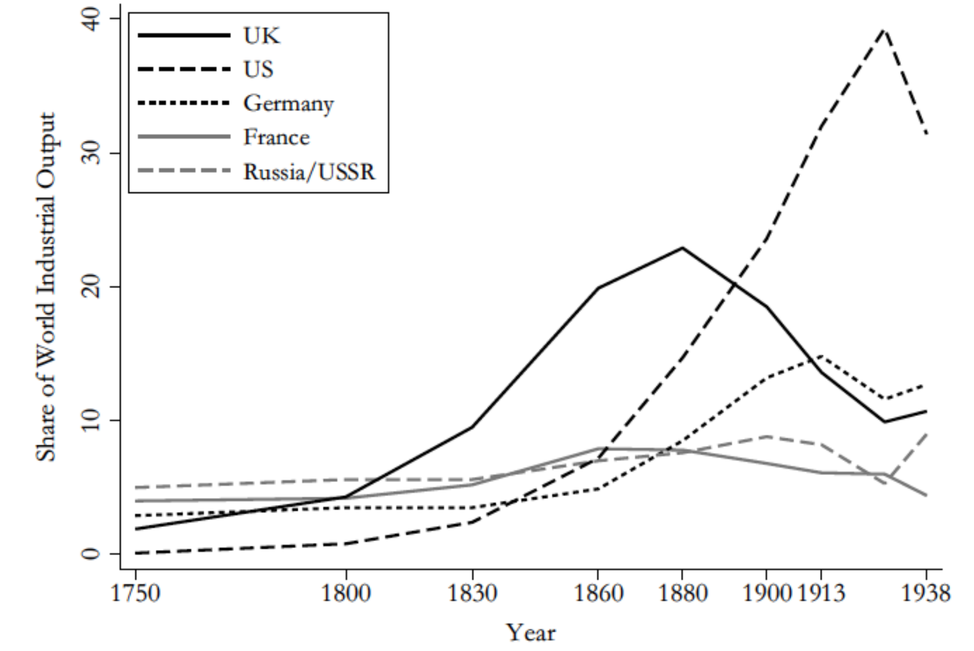


Figure 9.5 Shares of world industrial output, 1750–1938

Data source: Kennedy (1987, Tables 6, 18).

	1700-1820	1870-1913
Britain	0.34	1.01
Europe	0.15[1]	1.31[2]
United States	0.73	1.80

[1]Total Western Europe (Maddison (2000) p. 264).
[2]Average of Western European 16 (Maddison, 2003).

The Economic Rise of the USA

- The Ten Largest Industries in the US in 1860 and 1910

Table 1.14. *The sectoral distribution of GNP 1840–1900 (percentages)*

	1840	1850	1860	1870	1880	1890	1900
Agriculture	41	35	35	33	28	19	18
Manufacturing, mining, and hand trades	17	22	22	24	25	30	31
Transportation and public utilities	7	4	6	6	8	9	9
Commerce and all other private business	23	26	26	26	29	32	32
Government and education	2	2	2	2	2	3	3
Shelter	10	11	9	9	8	7	7

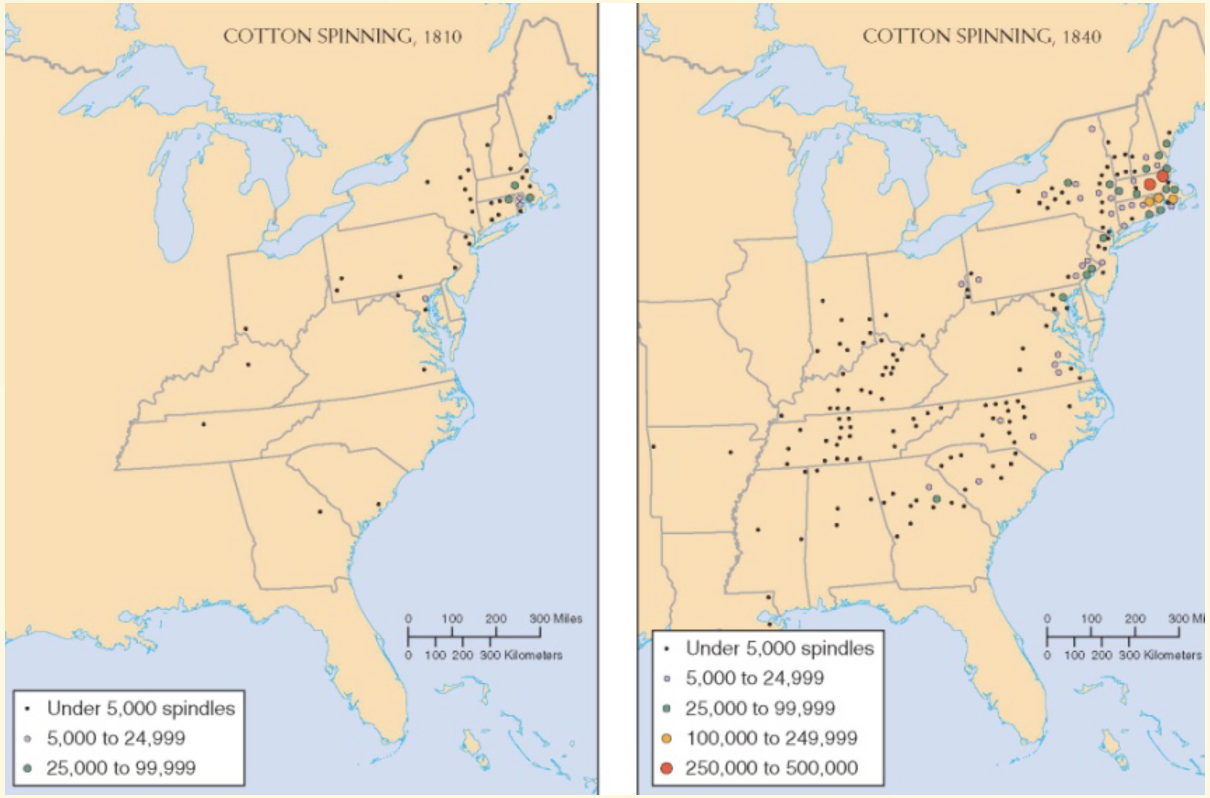
Note: The agricultural estimates include land clearing, breaking, and fencing as well as home manufacturing. The estimates, however, are understatements. If they were corrected, the share of agriculture in GNP would decline more pronouncedly over time than do the figures in this table. Mining excludes the important precious metals mining industries.
Source: Gallman "United States Capital Stock," table 4.8. These are estimates of sectoral value added. "Commerce" includes only the trade in final goods.

The Economic Rise of the USA

- The First Industrial Revolution (1760-1830)
- Began in England, spread to USA & Belgium in 1820s, Germany in the 1840s-1850s
- Key technologies: cotton textiles, steam engine, iron & steel
- Industrialization began in New England
- The Beverly Cotton Manufactory in Massachusetts in 1787
- Slater Mill in Rhode Island in 1790
- Waltham Mill in Massachusetts in 1814
- Timber, water, coal, iron, copper were abundant
- Lowell's Textile Mill in Massachusetts

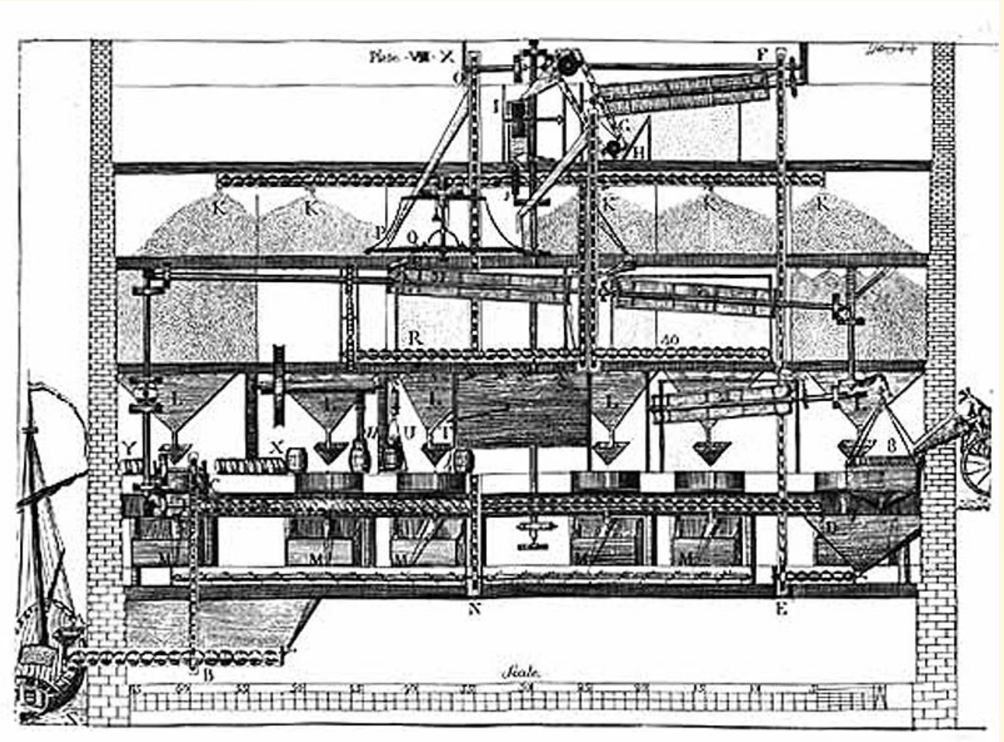
Cotton and Textiles

- Industrialization in the first half of the 19th century centered on cotton
- America was a technological follower
- Rapid growth was possible due to abundance of cheap factor inputs



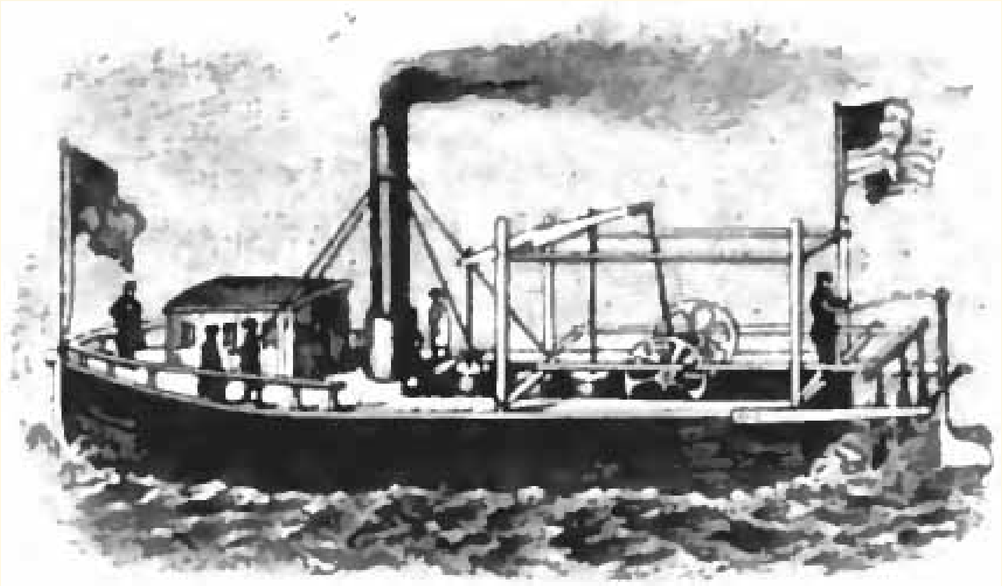
The Diffusion of Technology from Britain

- Recall Allen (2009): factor prices determine patterns of innovation
- Countries with factor prices similar to Britain should industrialize next
- North America had expensive labor and abundant natural resources. High cost of labor provided the incentive to innovate
- e.g. first automatic flour mill invented by Oliver Evans in 1782
- Belgium, the first European country to industrialize, had abundant coal



The Diffusion of Technology from Britain

- Mokyr (2009, 2016): a culture of growth or the Industrial Enlightenment was critical for the onset of economic growth
- America and Britain shared a common language and culture
- Attempts to restrict spread of IR technologies proved impossible
- Economic growth diffused first to Northern European societies that had participated in the Enlightenment and the Scientific Revolution

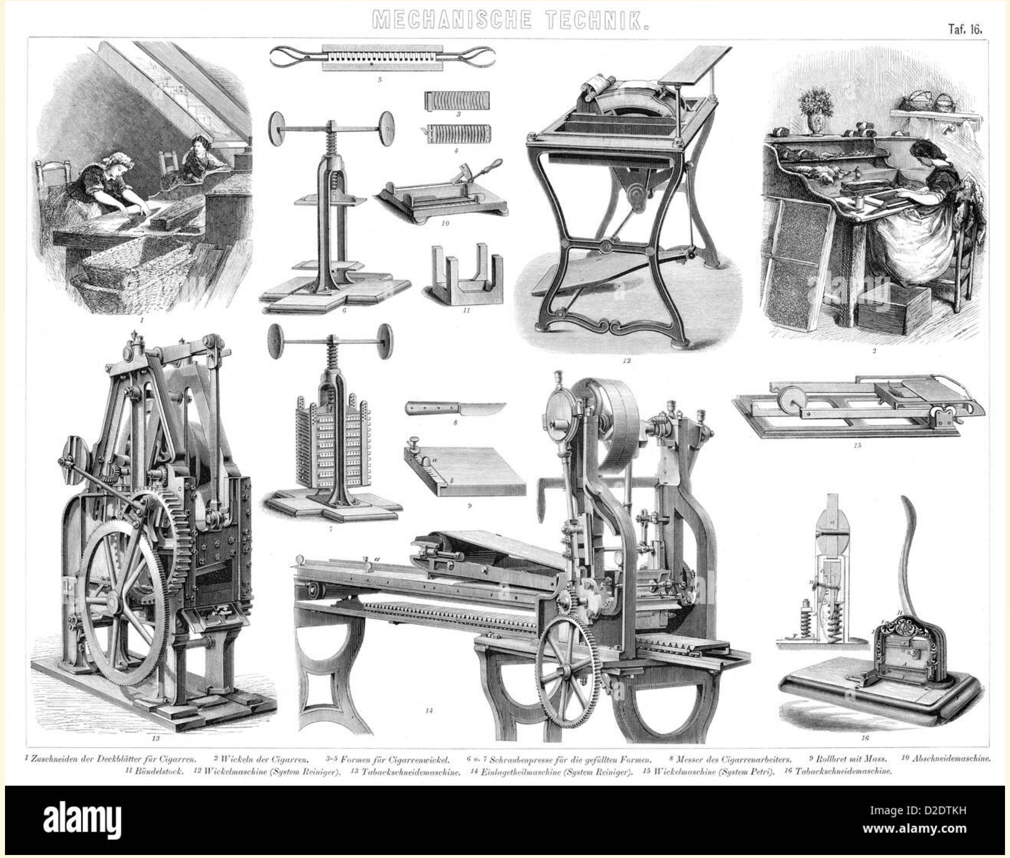


The Rise of the United States

- America's growth in the 19th century was also on the extensive margin
- Vast swathes of land were added and the population increased from around 2.5 million in 1776 to 76 million in 1900
- Immigration (especially from Ireland, Germany, Italy, and Eastern Europe) fueled this expansion
- This expansion in the size of the market complemented a distinctively "American" system of manufacturing
- US population and immigrant share, 1850-2010

The Second Industrial Revolution and American Economic Growth

- The American system of manufacturing centered on machine tools and the development of interchangeable parts
- A large market and homogeneous tastes encouraged product standardization
- Standardized products could be easily tweaked or improved upon
- Standardized products encouraged the standardization of other parts, and this made further innovation easier



The Soviet Detour I

- In the 20th century the Soviet Union was seen as pioneering an alternative growth model to that of the United States or Western Europe
- The Soviet Union achieved rapid industrialization in the 1930s (at vast human cost including millions of deaths in Ukraine due to famine) and fast growth for several decades
- This was interpreted as due to the success of economic planning

The Soviet Detour II

- Russia under the Tsars was a rapidly growing and industrializing economy, on course to achieve rapid catch-up growth and industrialization prior to WWI
- Much of the early Soviet growth was recovery from the devastation the Russian economy experienced in WWI and during the Russian civil war
- This growth was largely achieved by mobilizing tremendous resources (labor, land, natural resources, and physical capital)

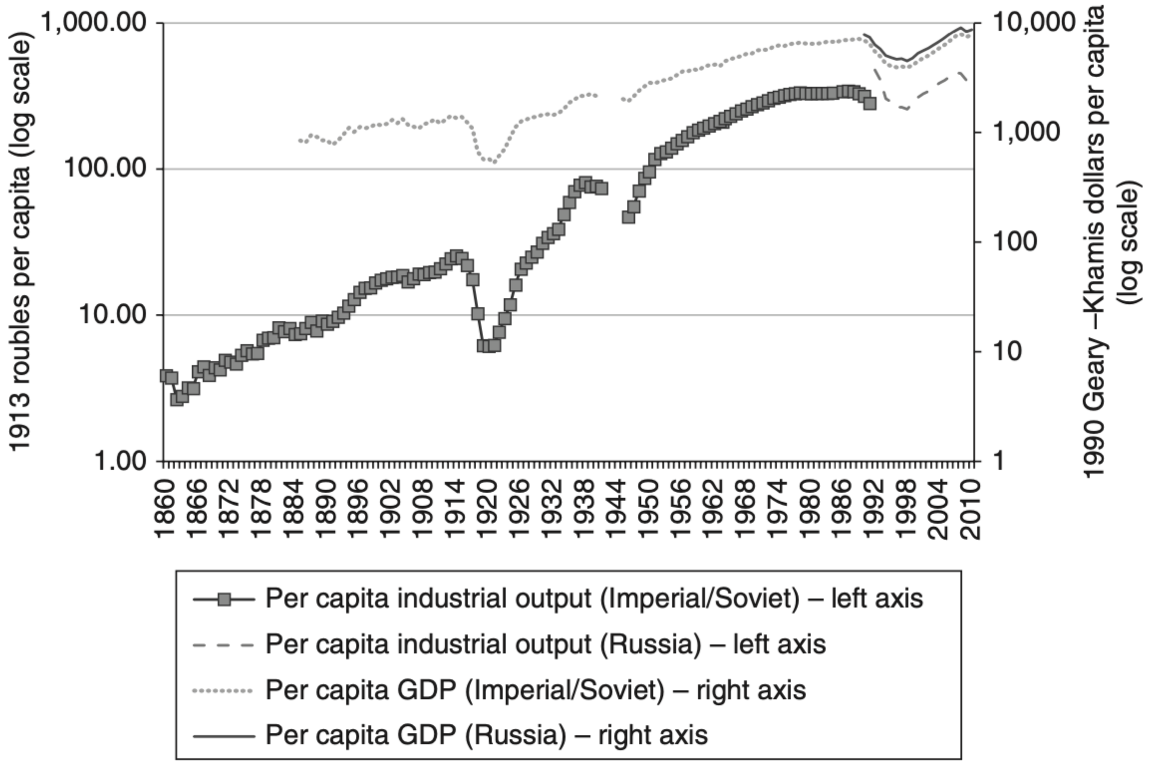
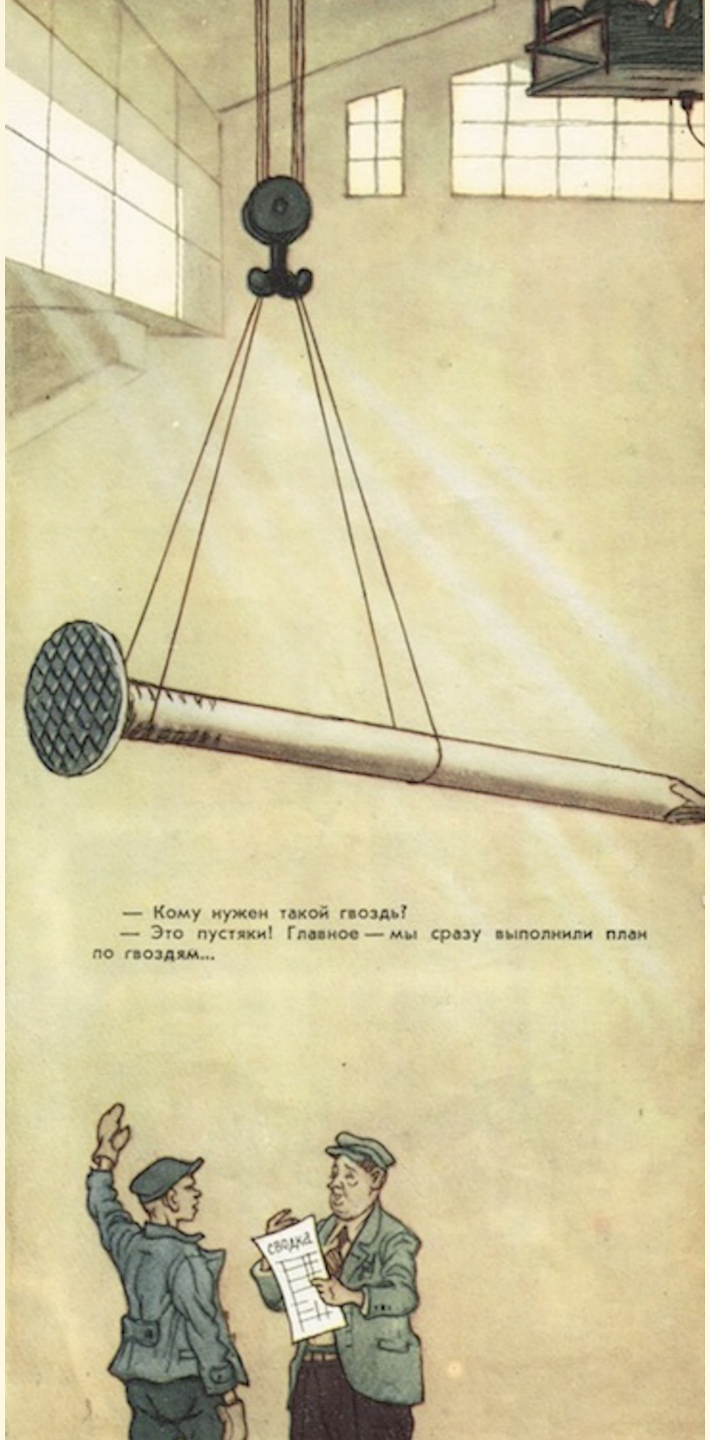


Fig. 3.1. Russian GDP and industrial output per capita

The Soviet Detour III

- Planning mobilized resources on a large scale but at tremendous costs
- Even during the early phase of rapid growth, inefficiencies and wastage were legendary (Gregory and Harrison, 2005).
- Centralization under a dictator produced chaos. Stalin spent time making seemingly trivial decisions, including setting price of metro tickets. Corruption and rent-seeking endemic
- Reforms after Stalin. But USSR struggled to provide basic consumer goods



Allen's "Standard Model" of Catch-up Growth

- Deliberate policy recipe used by 19th-century late industrializers (US, Germany) to catch up to Britain.
- **1. Internal Market Integration:** Building transport infrastructure (railways/canals).
- **2. External Protection:** Tariffs to protect nascent domestic industries from British competition.
- **3. Financial Institutions:** Creating investment banks to stabilize currency and provide industrial capital.
- **4. Mass Education:** Promoting schooling to facilitate technology adoption.

Applying the Standard Model

- **Success stories:** The United States and Germany successfully applied these policies to achieve rapid industrialization.
- **Mixed results:** Other nations, such as Imperial Russia, Japan, and Latin American countries, attempted the model with varying degrees of success.
- **Big Push:** In later stages of development, some countries shifted from this standard set of policies toward what Allen calls "Big Push" industrialization.

Allen's Revisionist View: The "Big Push"

- Argues that the Soviet "Big Push" model was a highly successful strategy for achieving rapid structural transformation.
- **Challenges the Tsarist Counterfactual:** Tsarist Russia was unlikely to achieve the same rapid transformation through market mechanisms alone.
- **Key Drivers:**
  - Massive reallocation of labor from low-productivity agriculture to high-productivity industry.
  - State-directed capital accumulation.
  - Demographic transition and emphasis on education.

Re-evaluating the Soviet Standard of Living

- Contrasts with the pessimistic view of purely falling living standards and famines.
- Data shows improvements in per capita consumption and quality-of-life indicators for both urban and rural populations by the late 1930s.
- **Nuance:** Distinguishes between the economic success of the industrialization strategy and the brutal political/human costs of forced collectivization.